PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

EdNote's E-Learning Website

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by:

Abhishek (2210991158)

Abhay (2210991140)

Aditya Sharma (2210991200)

Ayushman Bansal (2210991431)

Supervised By:

Dr Ajay Kumar

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY

CHITKARA UNIVERSITY, PUNJAB, INDIA

CONTENTS

S.No.	Title	Page No.
1.	Declaration	**3**
2.	Acknowledgement	**4**
3.	Abstract	**5**
4.	Introduction	**6**
5.	Methodology	**7**
6.	Tools and Technology	**8**
7.	Key Features	**9**
8.	Conclusion	**10**
9.	Future Scope	**11**
10.	Project Snapshots	**12**
11.	References	**16**

DECLARATION

We hereby certify that the work which is being presented in the project report entitled **"EdNote's E-Learning Website"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of **Dr Ajay Kumar**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Date: 30/04/2026

Abhishek
2210991158
Abhay
2210991140
Aditya Sharma
2210991200
Ayushman Bansal
2210991431

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr Ajay Kumar
Mentor
Department of Computer Science and Engineering
Chitkara University Institute of Engineering and Technology,
Chitkara University, Punjab, India

ACKNOWLEDGEMENT

It is our pleasure to be indebted to various people, who directly or indirectly contributed in the development of this work and who influenced our thinking, behaviour and acts during the course of study.

We express our sincere gratitude to all for providing us an opportunity to undergo Industry Oriented Hands On Experience as the part of the curriculum.

We are thankful to **Dr. Ajay Kumar** for his support, cooperation, and motivation provided to us during the training for constant inspiration, presence, and blessings.

Lastly, we would like to thank the almighty and our parents for their moral support and friends with whom we shared our day-to-day experience and received lots of suggestions that improve our quality of work.

Abhishek	**Abhay**	**Aditya Sharma**	**Ayushman Bansal**
2210991158	**2210991140**	**2210991200**	**2210991431**

Abstract

With the rise of the digital education system, students are rapidly accessing and depending on online learning resources for their academic support and exam preparation.

However, due to the availability of various educational websites, many platforms provide either study-related material or tests for practice and lack personalized practice tools but rarely all in a single well-organized platform.

This project proposes the development of an integrated e-learning platform that combines learning features including courses, MCQs practice, smart notes, interactive flashcards, and performance analytics into a single platform.

By reducing the need to switch between multiple websites, this system enhances learning experience, improves consistency, and supports better outcomes for students.

Problem Statement

In today's world there are availability of numerous online learning platforms, students face difficulties due to the lack of a unified learning system that combines study materials, practice tools, and performance analytics in one place. Many platforms focus on having single features such as providing only MCQs, courses or notes, which make students to suffer a lot and to switch between multiple websites.

Therefore, there is a need to develop an integrated e-learning website that brings together learning resources, interactive practice tools, and performance analytics to enhance learning experience and maintaining consistent study habits.

Introduction

With the increasing use of online learning, many platforms provide study materials, video lectures, practice tests and other tools that support learning and exam preparation.
However, the current digital learning system is highly disintegrated like providing only one aspect of learning, such as providing notes, quizzes etc. This is time consuming and also reduces learning efficiency, making it difficult for students to stay consistent in their studies.
To resolve these challenges, there is need for a unified e-learning platform that integrates all features into a single platform. **EdNote's** offer comprehensive study materials, practice tools like MCQs, smart notes and interactive flashcards, and feature for performance analytics. By providing a centralized learning environment, the system aims to enhance learning experience, improve efficiency for better outcomes.

Methodology

The system follows Agile based Software Development Life Cycle (SDLC) to ensure flexibility and continuous improvement.

1. **Planning:**
    - Project objectives, features and requirements were defined.

2. **Requirement Analysis:**
    - Functional and Non-Functional

3. **System Design:**
    - Design database schema (MongoDB)
    - Define API structure

4. **Development:**
    - Frontend built using React
    - Backend using Node.js and Express
    - Integration with MongoDB

5. **Authentication:**
    - JWT-based login system

6. **Testing:**
    - Unit, Integration

Tools and Technology

1. MongoDB:
    - A NoSQL database to store and manage data.
    - Enables fast retrieval and dynamic schema support.

2. Express.js:
    - To create APIs and handle server-side logic.
    - Powers secure data exchange between frontend and database.

3. React:
    - To build dynamic and responsive user interface.
    - It enables fast rendering and smooth user experience.

4. Node.js:
    - A server-side runtime environment that executes JavaScript.

5. Authentication:
    - JWT ensures only authorized users can access protected routes.
    - It maintains session management.

Key Features

1. **MCQ's and Courses:** EdNote's provide a comprehensive collection of MCQ feature and also including courses part, helping to practice topics and learning to enhance knowledge.

2. **Smart Notes:** This feature simplify the study process with adding notes of particular topic, to pin important ones, to search within notes and deleting notes to make revision faster and more effective.

3. **Interactive Flashcards:** Make revision easy by using flashcards to support in recalling topics that help in retain information longer in very effective way.

4. **Analytics:** This feature provides a visual representation of score trend and topic-wise performance through charts. Helping users to get a clear view to identify areas to do better and improve accordingly.

Conclusion

EdNote's E-Learning Website is a well-structured e-learning platform that enhances learning through MCQ based practice, courses. It also includes smart notes, interactive flashcards and analytics features to help learners.

- Enhances learning efficiency and user engagement.
- Improves accessibility by providing a centralized online learning website.
- Supports comprehensive learning with combined features.
- Enables consistent practice, leading to improved performance.

Future Scope

- **AI Integration:** Use of AI will enhance learning and make it easier for user to ask for any question and receive instant support, which helps maintain their interest and focus on their goals.

- **Offline Study Mode:** Users will be able to download courses, notes, and flashcards to study without any internet connection.

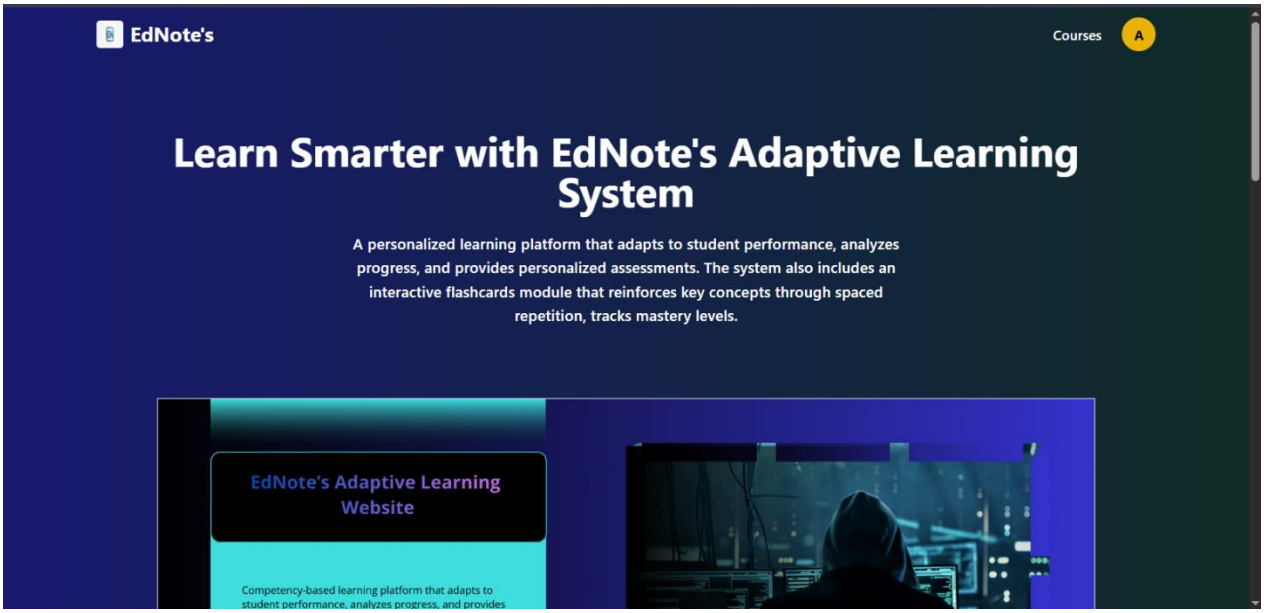
- **Certification:** After completing every module or courses, certificates will be issued to boost your career prospects.

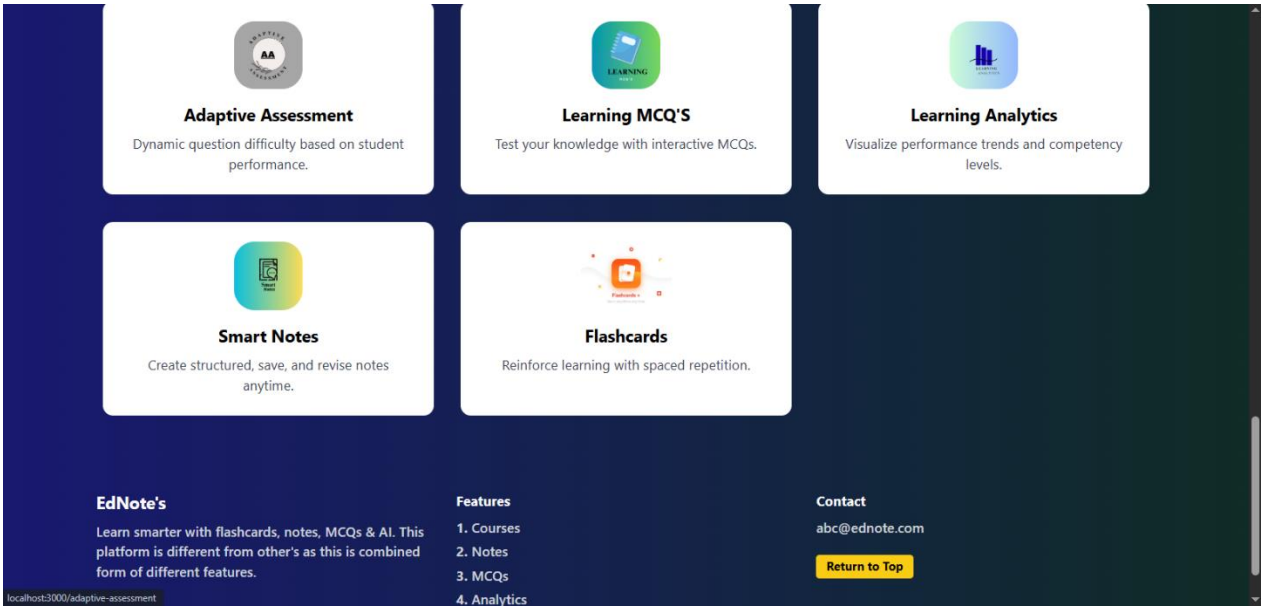
- **Mobile App:** Making the platform accessible anytime.

- **Paid Courses:** Users will be able to purchase courses through secure payment transactions.

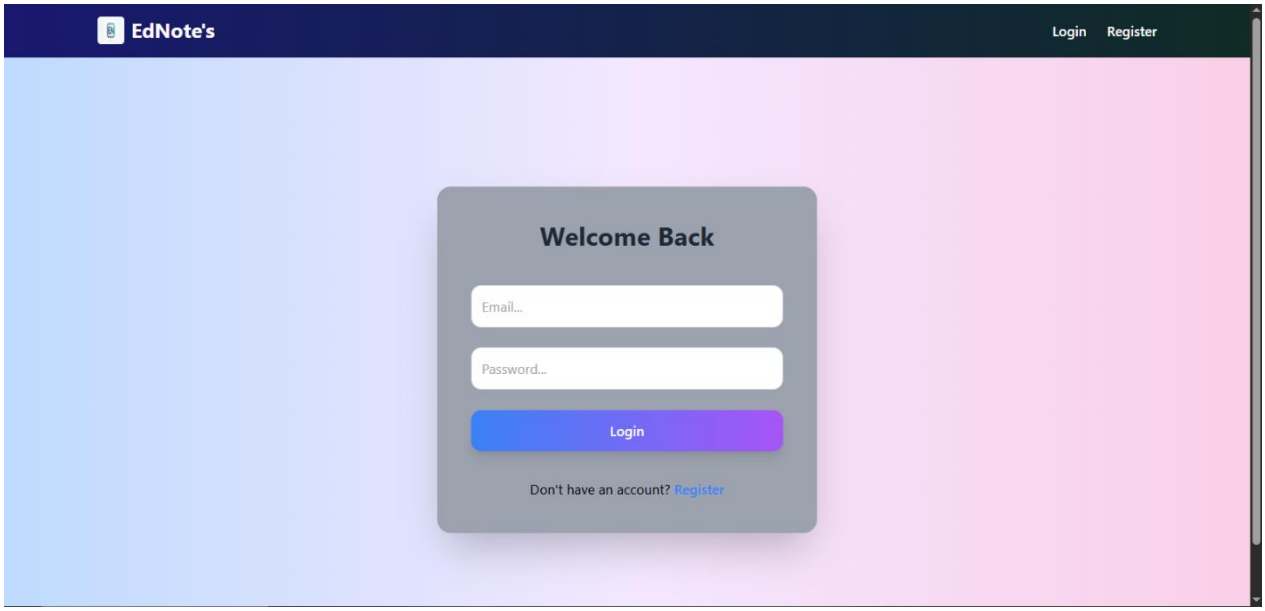
Project Snapshots

Home Page

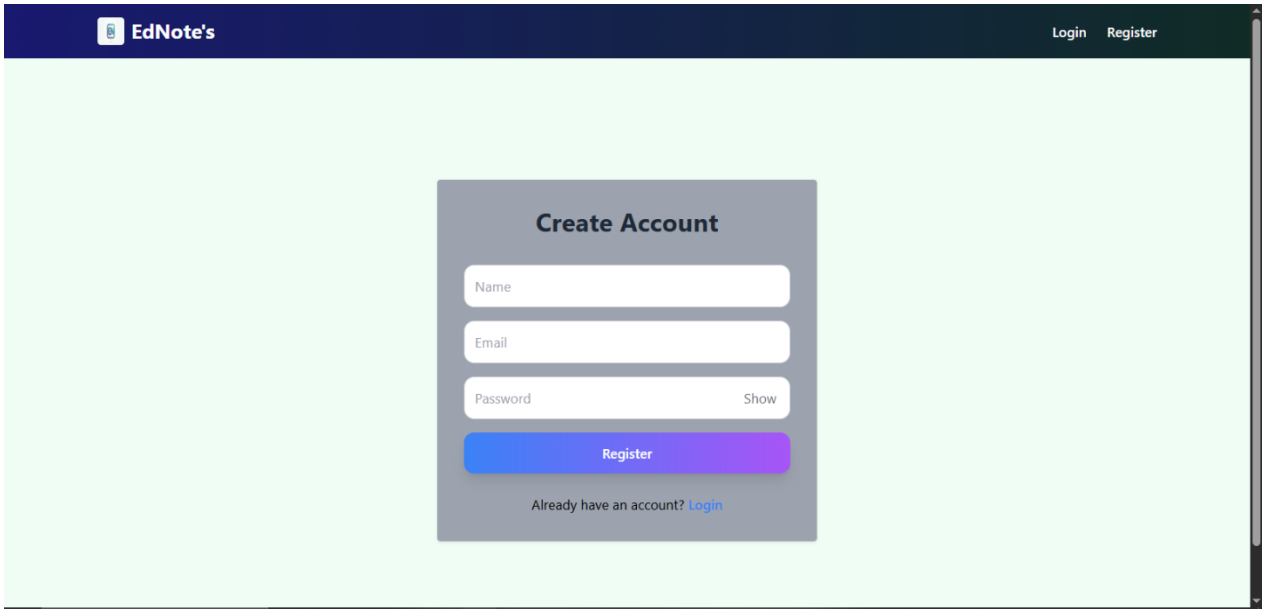




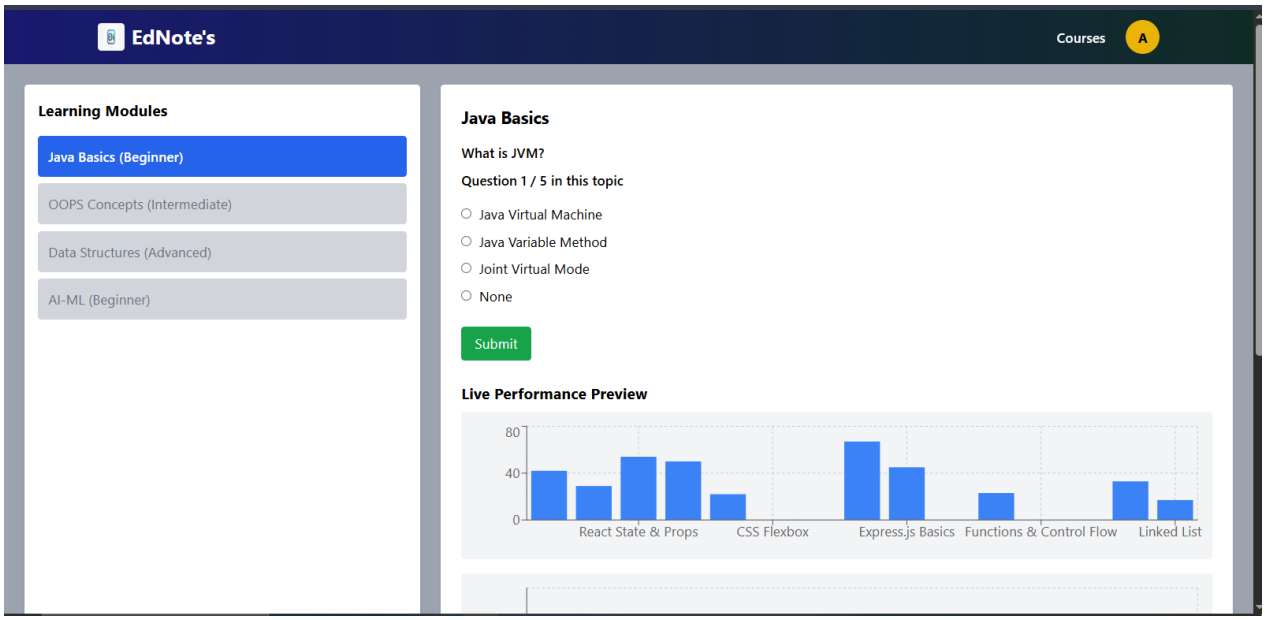
User Login



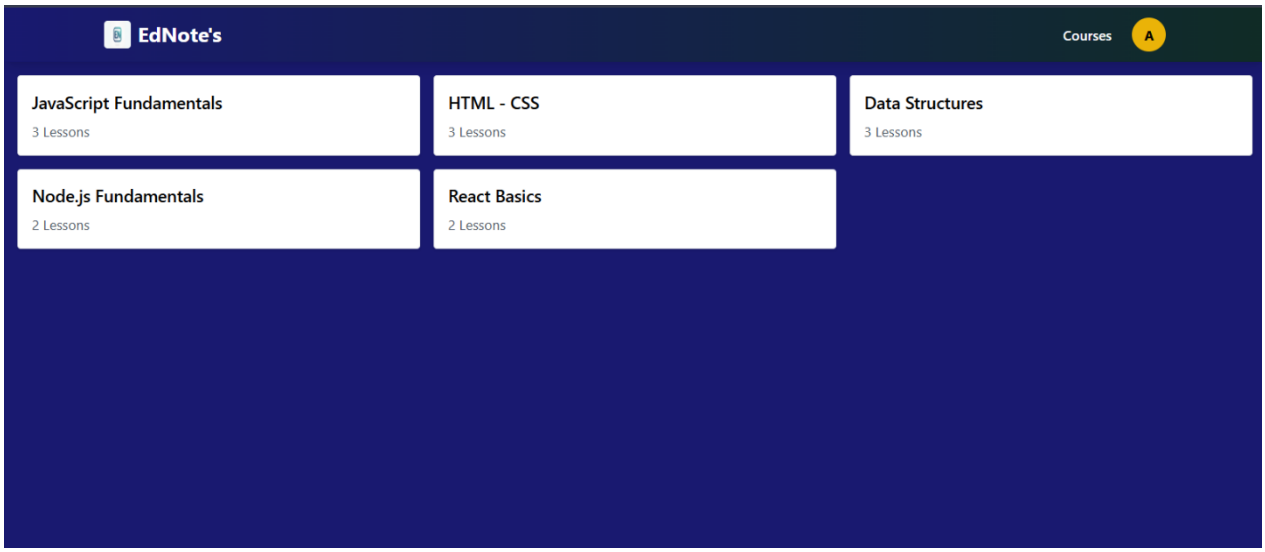
User Registration



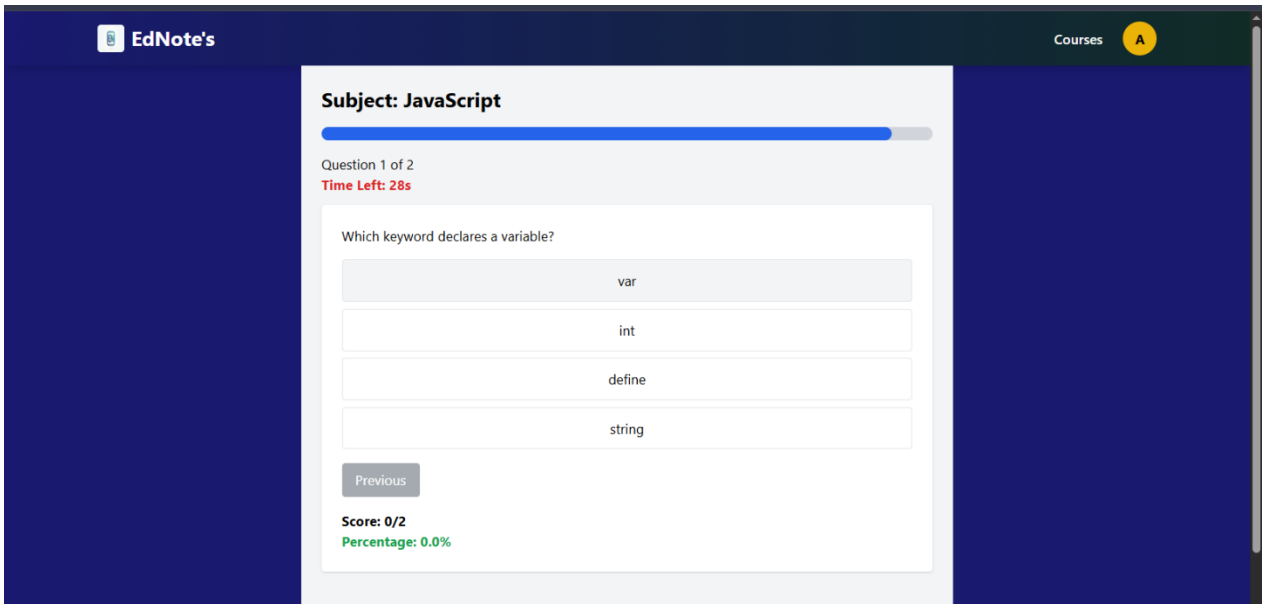
Adaptive Assessment



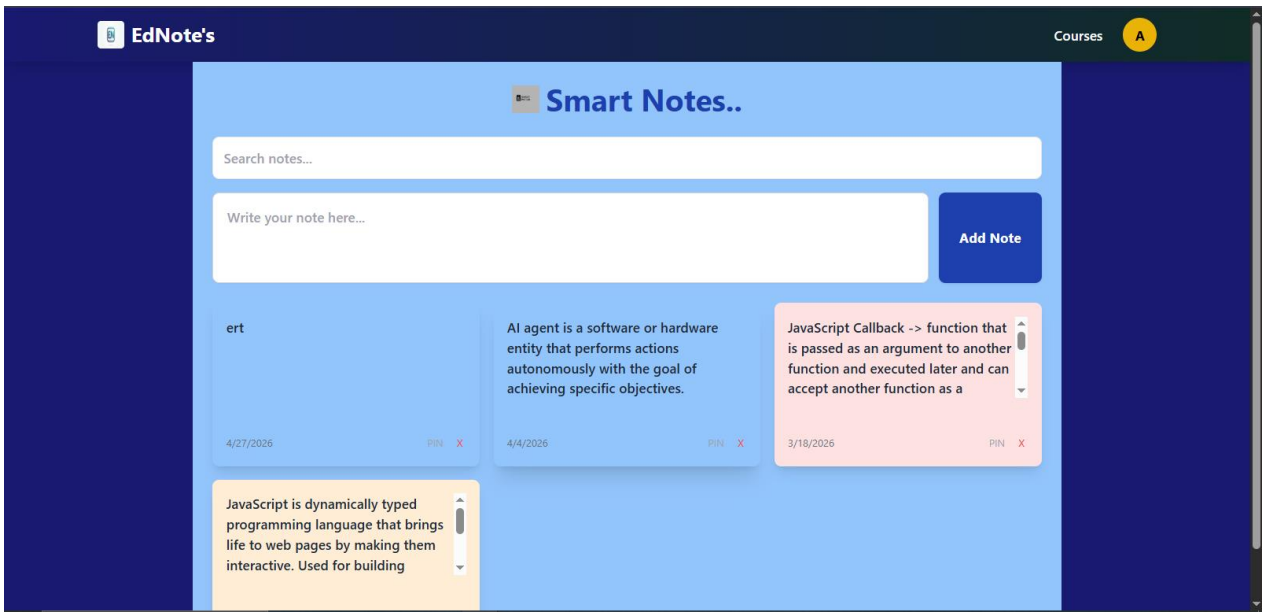
Courses



Learning MCQ



Smart Notes



References

- **https://www.mongodb.com**
- **https://expressjs.com**
- **https://reactjs.org**
- **https://nodejs.org**